

JKBOSE 2026

CLASS 11TH

CLASSIFICATION OF ELEMENTS AND PERIODICITY IN PROPERTIES

LECTURE-3

CHEMISTRY

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ON
TOP



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Topics To Covered

1

MODERN PERIODIC TABLE (INTRODUCTION).

REVISION

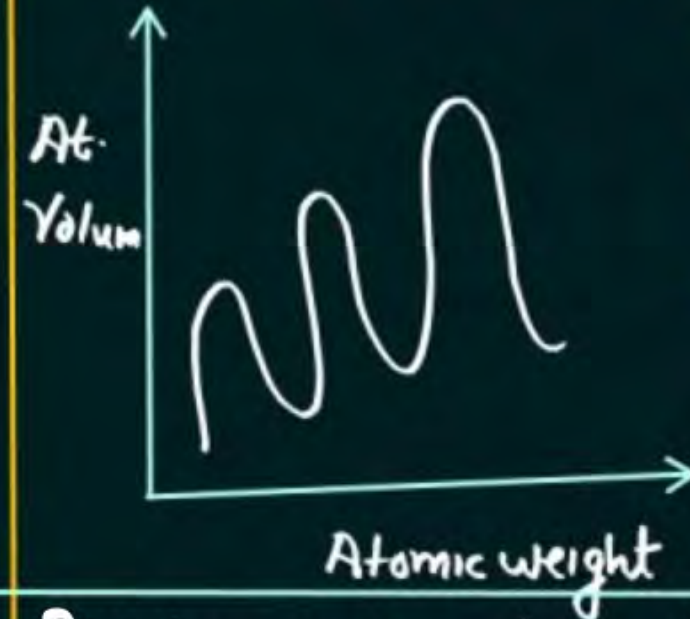
1) DOBEREINER TRIADS.

Li	Ca	Cl
Na	Sr	Br
K	Ba	I

2) NEWLANDS LAW OF OCTAVES.

Li	Be	B	C	N	O	F
Na	Mg	Al	Si	P	S	Cl
K	Ca					

3) LOTHAR MEYER



- Peak $\rightarrow$ Alkali Metals
- Descending $\rightarrow$ Alkaline earth metals
- Ascending $\rightarrow$ Halogens.

4) MENDELEEV'S PERIODIC TABLE.

PREDICTION

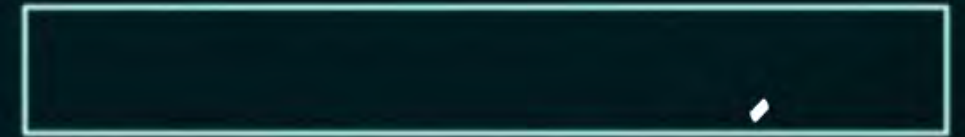
Eka-Al $\rightarrow$ Ga
Eka-Si $\rightarrow$ Ge

PROPERTIES.

$\rightarrow$ Groups $\rightarrow$ Eight
I, II, ..., VIII



PERIODS = SIX (1-6).



MERITS

- Systematic study
- Correction of Atomic masses.
- Prediction of new elements

DEMERITS

- POSITION OF Hydrogen.
- Anomalous pair

Ar (39.9)	K
Co (58.9)	Ni
- POSITIONS OF ISOTOPES.
- Cause of Periodicity
- Lanthanoids & Actinoids

MERITS OF MENDELEEV'S PERIODIC TABLE.

i) Systematic study of Elements $\Rightarrow$ He classified the elements with similar properties in a particular group. By knowing the property of one element of that group, we can easily remember the properties of all elements of that group.

ii) CORRECTION OF ATOMIC MASS $\Rightarrow$ He corrected the atomic mass of few elements like Be (13.5 u to 9 u), Li, Au etc.

iii) PREDICTION OF NEW ELEMENTS $\Rightarrow$

He predicted the properties of ^{few} undiscovered elements like Eka-Al (Ga), & Eka-Si (Ge).

DEFECTS OF MENDELEEV'S PERIODIC TABLE.

1). POSITION OF HYDROGEN $\rightarrow$ Few properties of hydrogen match with Alkali metals & few other properties match with halogen. but Hydrogen is placed in a group of Alkali metals.

2). ANOMALOUS PAIRS $\rightarrow$ Few pairs of elements were not placed in increasing order of atomic weight. e.g.

• Ar (39.9) & K (39.1 u)

• Co (58.9) & Ni (58.6 u)

DMITRI IVANOVICH
MENDELEEV. (RUSSIAN).

3). POSITION OF ISOTOPES $\rightarrow$

Isotopes are elements with same atomic number and different atomic mass. Therefore they should be placed in the periodic table, but this is not done.

4). Cause of Periodicity $\rightarrow$ Mendeleev was unable to explain the cause of periodicity of elements.

5) POSITION OF LANTHANIDES AND ACTINIDES.

Lanthanides and Actinides are 28 elements. were not given any position in Mendeleev's periodic table..

MODERN PERIODIC LAW.

In, 1912. MOSELEY observed that X-rays emitted from elements when high speed electron are bombarded on them. The square root of frequencies of elements were found to be directly proportional to the atomic number of Elements. i.e.

$$\sqrt{\nu} \propto Z$$

He concluded that, instead of atomic weight, atomic number is the fundamental property of elements. And proposed a Modern Periodic Law.

It states that
"Physical & chemical properties of elements are periodic function of atomic number."

MODERN PERIODIC LAW OR LONG FORM OF PERIODIC TABLE



Features of the Long Form of the Periodic Table (Bohr's Table):

This table is based on how electrons are arranged in atoms (electronic configuration). It is also called Bohr's table because it follows Bohr's model of the atom.

Main Features:

1. Groups (Vertical Columns):

There are 18 vertical columns in the table, called groups or families.

These groups are numbered from 1 to 18.

2. Periods (Horizontal Rows):

The table has 7 horizontal rows, called periods, numbered from 1 to 7 (top to bottom).

The period number shows the highest energy level (shell) of electrons in an atom of the element.

3. Number of Elements in Periods:

- 1st period: 2 elements (Very short Period)
- 2nd & 3rd periods: 8 elements each (Short Period)
- 4th & 5th periods: 18 elements each (Long Period)
- 6th & 7th periods: 32 elements each (Longest Period)

1	Period	→	2 elements
2	→		8
3	→		8
4	→		18
5	→		18
6	→		32
7	→		32

2, 8, 8, 18, 18, 32, 32.

↓
Magic numbers.

GROUP	1	2											13	14	15	16	17	18
PERIOD	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1	H Hydrogen 1.008																	He Helium 4.003
2	Li Lithium 6.94	Be Beryllium 9.012											B Boron 10.81	C Carbon 12.01	N Nitrogen 14.01	O Oxygen 16.00	F Fluorine 19.00	Ne Neon 20.18
3	Na Sodium 22.99	Mg Magnesium 24.31											Al Aluminum 26.98	Si Silicon 28.09	P Phosphorus 30.97	S Sulfur 32.06	Cl Chlorine 35.45	Ar Argon 39.95
4	K Potassium 39.10	Ca Calcium 40.08	Sc Scandium 44.96	Ti Titanium 47.88	V Vanadium 50.94	Cr Chromium 52.00	Mn Manganese 54.94	Fe Iron 55.85	Co Cobalt 58.93	Ni Nickel 58.69	Cu Copper 63.55	Zn Zinc 65.39	Ga Gallium 69.72	Ge Germanium 72.64	As Arsenic 74.92	Se Selenium 78.96	Br Bromine 79.90	Kr Krypton 83.79
5	Rb Rubidium 85.47	Sr Strontium 87.62	Y Yttrium 88.91	Zr Zirconium 91.22	Nb Niobium 92.91	Mo Molybdenum 95.96	Tc Technetium (98)	Ru Ruthenium 101.1	Rh Rhodium 102.9	Pd Palladium 106.4	Ag Silver 107.9	Cd Cadmium 112.4	In Indium 114.8	Sn Tin 118.7	Sb Antimony 121.8	Te Tellurium 127.6	I Iodine 126.9	Xe Xenon 131.3
6	Cs Cesium 132.9	Ba Barium 137.3	57-71 Lanthanides	Hf Hafnium 178.5	Ta Tantalum 180.9	W Tungsten 183.8	Re Rhenium 186.2	Os Osmium 190.2	Ir Iridium 192.2	Pt Platinum 195.1	Au Gold 197.0	Hg Mercury 200.5	Tl Thallium 204.38	Pb Lead 207.2	Bi Bismuth 209.0	Po Polonium (209)	At Astatine (210)	Rn Radon (222)
7	Fr Francium (223)	Ra Radium (226)	89-103 Actinides	Rf Rutherfordium (261)	Db Dubnium (268)	Sg Seaborgium (271)	Bh Bohrium (270)	Hs Hassium (277)	Mt Meitnerium (276)	Ds Darmstadtium (281)	Rg Roentgenium (280)	Cn Copernicium (285)	Nh Nihonium (284)	Fl Flerovium (289)	Mc Moscovium (288)	Lv Livermorium (293)	Ts Tennessine (294)	Og Oganesson (294)

78 — Atomic Number

Pt — Symbol

Platinum — Name

195.1 — Average Atomic Mass

57 La Lanthanum 138.9	58 Ce Cerium 140.1	59 Pr Praseodymium 140.9	60 Nd Neodymium 144.2	61 Pm Promethium (145)	62 Sm Samarium 150.4	63 Eu Europium 152.0	64 Gd Gadolinium 157.2	65 Tb Terbium 158.9	66 Dy Dysprosium 162.5	67 Ho Holmium 164.9	68 Er Erbium 167.3	69 Tm Thulium 168.9	70 Yb Ytterbium 173.0	71 Lu Lutetium 175.0
89 Ac Actinium (227)	90 Th Thorium 232.0	91 Pa Protactinium 231.0	92 U Uranium 238.0	93 Np Neptunium (237)	94 Pu Plutonium (244)	95 Am Americium (243)	96 Cm Curium (247)	97 Bk Berkelium (247)	98 Cf Californium (251)	99 Es Einsteinium (252)	100 Fm Fermium (257)	101 Md Mendelevium (258)	102 No Nobelium (259)	103 Lr Lawrencium (262)

QUESTION

Successive filling of $3s$ and $3p$ orbitals give rise to the third period. The number of elements present in this period are

1 2

2 4

3 6

4 8

$$1 \rightarrow 2$$

$$2+3 \rightarrow 8, 8$$

$$4+5 \rightarrow 18, 18$$

$$6+7 \rightarrow 32, 32$$

Q. Which of the following triad is present in NEWLANDS CLASSIFICATION.

a) Ca, Sr, Ba

b) Li, Na, K

c) Cl, Br, I

d) Fe, Co, Ni

Q. According to MOSLEY which of the following is fundamental property for classification of Elements.

a) Atomic Number

b) Atomic mass.

c) Atomic Volume

d) Mass Number.

Latter sawal

Q. Modern periodic table consists of

- a) 7 groups, 18 Period.
- b) 17 group, 8 Period
- c) 18 Group, 7 Periods
- d) 8 group, 7 Periods

GROUP



1 $\rightarrow$ 18

18

PERIOD



1-7
7

Q. Seventh Period involves the filling of

a) 7s 6d 7p

b) 7s 4f 5d 7p.

c) 7s 5f 6d 7p

d) 7s 6f 5d 7p.

REVISION KAR K MI SONA.

Thank
You